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Decision-focused linear pooling for probabilistic forecast combination

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ABSTRACT

In real-world settings, decision-makers often have access to multiple forecasts for the same unknown quantity. Combining different forecasts has long been known to improve forecast quality, as measured by scoring rules in the case of probabilistic forecasting. However, improved forecast quality does not always translate into better decisions in a downstream problem that utilizes the resultant combined forecast as input. To this end, this work proposes a novel probabilistic forecast combination approach that accounts for the downstream stochastic optimization problem by which the decisions will be made. We propose a linear pool of probabilistic forecasts where the respective weights are learned by minimizing the expected decision cost of the induced combination, which we formulate as a nested optimization problem. Two methods are proposed for its solution: a gradient-based method that utilizes differential optimization layers, and a performance-based weighting method. The proposed decision-focused combination approach is validated in two integral problems associated with renewable energy integration in low-carbon power systems and compared against well-established combination methods. Namely, we examine an electricity market trading problem under stochastic solar production and a grid scheduling problem under stochastic wind production. The results illustrate that the proposed approach leads to lower expected downstream costs, while optimizing for forecast quality when estimating linear pool weights does not always translate into better decisions. Notably, optimizing for a combination of downstream cost and an accuracy-oriented scoring rule consistently leads to better decisions while also improving forecast quality.

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1. Introduction

1.1. Context and motivation

Real-world decision-making problems typically involve uncertain parameters corresponding to quantities, such as product demand or market prices, which are unknown at decision time. Classical stochastic optimization (Birge & Louveaux, 2011) assumes that these uncertain parameters follow a known distribution. As only observational data are available, uncertainty distributions are estimated through short-term probabilistic forecasting, which becomes a critical input in decision-making pipelines.

In many real-world settings, decision-makers have multiple expert forecasts for the same uncertain parameter. This setting is common, for instance, in power and energy systems with a high penetration of variable renewable energy sources. To ensure a reliable and cost-effective operation, grid operators further employ domain experts, such as meteorologists and power system engineers, who, given additional available information such as

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current weather conditions, select or combine individual forecasts (Motley, 2023). The resultant combined forecasts are subsequently inputted within operational workflows, such as market clearing algorithms and production cost models.

Ultimately, the decision-maker combines available forecasts to derive a downstream decision. Several recent works have illustrated that increased forecast quality (either in a point or probabilistic sense) does not always translate into better decisions (Mandi et al., 2024). This effect becomes even more pronounced in risk-critical infrastructures, such as power systems, where forecast errors induce highly asymmetrical costs (Morales, Muñoz, & Pineda, 2023). Hence, it is critical to go beyond forecast quality and explicitly consider forecast *value* within a decision-making pipeline when combining available forecasts.

1.2. Literature review

Probabilistic forecasting. Gneiting, Balabdaoui, and Raftery (2007) posit that probabilistic forecasting aims to maximize sharpness subject to calibration. Forecast quality is evaluated based on so-called *proper scoring rules* (Gneiting & Raftery, 2007), which compare probabilistic forecasts against realizations of a random target variable and elicit an honest uncertainty estimation by the expert. As the decision-maker might not be uniformly interested in forecast quality across the whole distribution, weighted scoring rules can be utilized to evaluate probabilistic forecasts with emphasis on specific regions of interest (Gneiting & Ranjan, 2011).

Contextual stochastic optimization. Contextual stochastic optimization refers to optimization problems where uncertain parameters are associated with some contextual information (or *features*) and is closely related to probabilistic forecasting. For instance, renewable energy production is associated with the weather, market-clearing prices depending on demand, etc. The goal here is to minimize an expected decision cost given a realization of associated feature data (Bertsimas & Kallus, 2020). The standard two-step approach involves forecasting uncertain parameters and solving an optimization problem. Recently, there has been a growing interest in integrating forecasting and optimization to generate forecasts that lead to better downstream decisions, termed *decision-focused learning*. One pertinent idea is embedding the objective of the downstream problem within the learning process (Elmachtoub & Grigas, 2022). Mandi et al. (2024) provide a comprehensive overview of decision-focused learning for linear programming problems with uncertain cost coefficients, which only require *point* forecasts as input. Nonetheless, in stochastic optimization problems, the uncertainty may have a nonlinear effect on the induced cost, necessitating *probabilistic* forecasts as input instead. Probabilistic forecasts can be implemented with a variety of methods, including parametric distributional models, non-parametric machine learning models (Bertsimas & Kallus, 2020), conformal prediction (Angelopoulos et al., 2023), or multi-label classification. Similarly to the point forecasting case, there has been a growing interest in decision-focused probabilistic forecasting (Donti, Amos, & Kolter, 2017; Kallus & Mao, 2023; Qi, Grigas, & Shen, 2023; Stratigakos, Camal, Michiorri, & Kariniotakis, 2022). A perfect probabilistic forecast, i.e., one that perfectly models the true probability distribution, would always be preferred by decision-makers, regardless of their downstream objective (Diebold, Gunther, & Tay, 1998). However, a degree of model misspecification and, subsequently, forecast error is unavoidable, which motivates a decision-focused approach that aligns the estimation of the probabilistic model with the downstream objective, implicitly accounting for different regions of interest (Gneiting & Ranjan, 2011). Such an approach regularly outperforms the standard "forecast, then optimize" approach in several problems across industries, such as low-carbon power and energy systems (Chen, Yang, Liu, & Wu, 2022).

Combining probabilistic forecasts. Forecast combination has long been known to improve forecast quality over individual (or *component*) forecasts (Wang, Hyndman, Li, & Kang, 2023), both in point and probabilistic cases. The prevalent method for combining forecast distributions is the linear pool, which dates back at least to Stone (1961), Winkler (1968), where each expert is assigned a weight that signifies their forecast skill. Krüger (2013) shows that a linear pool is always more accurate for several proper scoring rules than the worst-performing component forecast, both in expectation and for a single observation. In a decision-making context, Stone (1961) shows that the expected utility (cost) of the decision obtained from the linear pool is higher (lower) than the worst-case utility (cost) of the component forecasts, i.e., the decision selected by using a component forecast. In other words, the linear pool will do necessarily better than the worst component forecast. Forecasting models are often wrong, so the linear pool effectively hedges against model misspecification. Linear pooling is typically used to combine or average probabilities in the form of cumulative density functions. Raftery, Gneiting, Balabdaoui, and Polakowski (2005) linearly pool weather ensembles leveraging Bayesian model averaging (Hoeting, Madigan, Raftery, & Volinsky, 1999) to generate probabilistic forecasts. Lichtendahl, Grushka-Cockayne, and Winkler (2013) examine conditions under which it is favorable to linearly pool quantile functions rather than probabilities. Kapetanios, Mitchell, Price, and Fawcett (2015) generalize linear pools by allowing outcome-dependent weights. Gneiting and Ranjan (2013) highlight some shortcomings of linear pooling, namely, a linear pool of calibrated forecasts may be uncalibrated and underdispersed, and further propose nonlinear pooling based on the Beta transformation. In many practical settings, however, individual probabilistic forecasts tend to be underdispersed; hence, linear pooling improves overall forecast quality.

Learning linear pool weights. Despite the shortcomings highlighted by Gneiting and Ranjan (2013), linear pooling performs well in practice and is the most popular combination method. Given the robustness of simple approaches, Wang et al. (2023) posit that forecast combinations should aim to be "sophisticatedly simple". The

question thus arises: *How to properly tune the weights of a linear pool?* An equally weighted linear pool, termed *ordinary linear pool*, has been proven fairly robust and is generally hard to beat in empirical evaluations. Geweke and Amisano (2011) estimate weights that maximize the logarithmic score. Papayiannis and Yannacopoulos (2018) show that quantile averaging is a special case of the Wasserstein barycenter and estimate weights that minimize the Wasserstein distance of the combined forecast and observed data. Recent works focus on minimizing the induced forecast's continuous ranked probability score (CRPS), which is the gold standard in probabilistic forecast evaluation. Thorey, Mallet, and Baudin (2017) minimize a bias-corrected CRPS for ensemble forecasting and Thorey, Chaussin, and Mallet (2018) further extend this work to probabilistic solar production forecasting. van der Meer, Pinson, Camal, and Kariniotakis (2024) develop an online algorithm to adaptively minimize the CRPS of a Beta-transformed linear pool of renewable production forecasts. Berrisch and Ziel (2023) observe differences in forecast quality across various distribution regions and propose a pointwise quantile combination minimizing the aggregate CRPS. Krannichfeldt, Wang, Zufferey, and Hug (2022) average quantiles by minimizing the quantile score, which approximates CRPS, and updating the combination weights only when the loss exceeds a predetermined threshold.

Nonetheless, the forecast quality versus value gap also appears in forecast combinations. Nitka and Weron (2023) examine the efficacy of CRPS-based combination weights when trading in electricity markets under uncertain prices. Their extensive empirical study finds that, although CRPS-based weights lead to more accurate forecasts, their value is negligible compared to an ordinary linear pool. Only a few recent works account for the importance of the downstream optimization problem when combining forecasts. Tallman and West (2024a) propose a Bayesian framework for model combination based on expected decision outcomes, termed Bayesian Predictive Decision Synthesis (BPDS). BPDS utilizes entropic tilting and multiple scoring rules to assign linear pooling weights based on historical decision quality. In a subsequent work, Tallman and West (2024b) present a detailed application of BPDS to portfolio optimization.

1.3. Aim and contribution

In this work, we develop a decision-focused approach to estimate linear pool weights for probabilistic forecasting. Given a data-driven, contextual stochastic optimization problem, we learn combination weights by minimizing the expected incurred decision cost. Contrary to previous work on decision-focused combination (Tallman & West, 2024a, 2024b), we consider constrained optimization problems, such as two-stage linear programming (LP) problems. We propose two solution methods for the resultant problem, namely, a differentiable optimization-based method (Agrawal, Barratt, & Boyd, 2021) and weighting component forecasts based on their historical, in-sample decision quality, which adapts the classic approach of Bates and Granger (1969) in a contextual optimization framework. Further, we extend our approach to a conditional combination setting, wherein the combination weights depend on additional features that become available before making decisions, which is of great practical interest and has not been previously addressed (Tallman & West, 2024a). Importantly, we make two key contributions: *(i)* we connect the weight estimation process with the downstream decision-making process while retaining a simple combination scheme, and *(ii)* account for additional information that might be available to the decision-maker. We validate the proposed combination approach in critical applications related to variable renewable energy sources integration in low-carbon power systems (Morales, Conejo, Madsen, Pinson, & Zugno, 2013). We examine an electricity market trading problem under stochastic solar production and a grid scheduling problem under stochastic wind production. The numerical results show that embedding the downstream problem by co-optimizing decision costs and forecasting quality consistently improves decision outcomes. Notably, optimizing for a combination of downstream cost and CRPS consistently leads to better decisions while maintaining high forecast quality.

The rest of the paper is organized as follows. Section 2 presents preliminaries, formulates the proposed combination approach, and develops the solution methods. Section 3 presents the results of the numerical experiments. Section 4 summarizes and discusses future work.

2. Methodology

This section introduces our notation and develops the proposed methodology. Section 2.1 introduces contextual stochastic optimization, probabilistic forecast evaluation, and forecast combination. Section 2.2 formulates the proposed decision-focused forecast combination approach, which is our main contribution. Finally, Section 2.3 develops two solution methodologies for the decision-focused forecast combination.

Notation. Uppercase letters denote random variables, lowercase letters denote realizations, and $\hat{\cdot}$ denotes forecasts. Bold font denotes vectors, and normal font denotes scalar quantities. Sets are denoted with calligraphic font, e.g., $\mathcal{S}$, and $|\mathcal{S}|$ denotes the cardinality (number of elements) of a set $\mathcal{S}$. The notation $[n]$ is used as a shorthand for $1, \dots, n$. Further, let

$$\Sigma_n = \{\mathbf{a} \in \mathbb{R}^n_+ \mid \mathbf{a}^\top \mathbf{1}_n = 1\}$$

be the standard $(n-1)$-dimensional probability simplex, where $\mathbf{1}_n$ is an n-sized vector of ones. Finally, let $\mathbb{I}(\cdot)$ be the indicator function.

2.1. Methodology preliminaries

2.1.1. Contextual stochastic optimization

We consider a contextual stochastic optimization problem given by

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}[c(\mathbf{z}; Y)|\mathbf{X} = \mathbf{x}_0], \tag{1}$$

where $Y \in \mathcal{Y}$ denotes the uncertain problem parameters, $\mathbf{X} \in \mathcal{X}$ denotes some associated contextual features, $\mathbf{x}_0$

denotes a realization of $\mathbf{X}$, $\mathbf{z}$ denotes the decision variables, $\mathcal{Z}$ denotes the convex set of feasible solutions, c denotes a convex cost function, and the expectation is taken with respect to (w.r.t.) the conditional distribution of Y given $\mathbf{X} = \mathbf{x}_0$. For instance, in inventory management, uncertainty Y may refer to demand data and $\mathbf{X}$ to prices or discount actions; in power and energy systems, Y may refer to stochastic renewable production and $\mathbf{X}$ to weather conditions; in portfolio optimization problems, Y may refer to the time series of stock returns and $\mathbf{X}$ to past returns or market conditions.

Further assume that the uncertain parameter Y is a discrete random variable with finite support $\mathcal{Y} \overset{\text{def}}{=} \{\xi_1, \ldots, \xi_K\}$, where K is the number of support locations. For any instance $\mathbf{x}_0 \in \mathcal{X}$, the true conditional distribution of Y is given by a discrete probability vector (or histogram) $\mathbf{p}(Y \mid \mathbf{x}_0) \in \Sigma_K$, where Σ_K is the $(K-1)$-dimensional probability simplex. The kth component of $\mathbf{p}(Y \mid \mathbf{x}_0)$ is defined as $p_k = \mathbb{P}(Y = \xi_k | \mathbf{x}_0)$, i.e., the conditional probability of $Y = \xi_k$. Problem (1) can be equivalently written as

$$\min_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}[c(\mathbf{z}; Y)|\mathbf{x}_0] = \min_{\mathbf{z}\in\mathcal{Z}} \sum_{k\in[K]} p_k c(\mathbf{z}; \xi_k). \tag{2}$$

A relevant example we consider is two-stage LP problems with fixed recourse and known cost coefficients, given by

$$\begin{aligned} \min_{\mathbf{z}\geq 0, \mathbf{u}_k \geq 0} \quad & \mathbf{c}^\top \mathbf{z} + \sum_{k\in[K]} p_k (\mathbf{q}^\top \mathbf{u}_k), \\ \text{s.t.} \quad & \mathbf{A}\mathbf{z} = \mathbf{b}, \\ & \mathbf{T}(\xi_k)\mathbf{z} + \mathbf{W}\mathbf{u}_k = \mathbf{h}(\xi_k), \qquad k \in [K], \end{aligned}$$

where $\mathbf{u}_k$ are the recourse actions for the kth scenario, $(\mathbf{A}, \mathbf{b}, \mathbf{W})$ are fixed parameters, $(\mathbf{T}, \mathbf{h})$ are parameters that depend on the realized uncertainty, and $(\mathbf{c}, \mathbf{q})$ are known cost vectors.

In real-world applications, instead of the true probability vector $\mathbf{p}(Y \mid \mathbf{x}_0)$, we have access to some training data, which are used to approximate (2) by estimating the conditional distribution of Y through probabilistic forecasting. Let us further assume that S experts provide the decision-maker with probabilistic forecasts $\{\hat{\mathbf{p}}^s\}_{s\in[S]}$ that model the conditional distribution of Y and that the decision-maker has collected a data set of N historical observations of $\mathcal{D} = \{(y_i, \hat{\mathbf{p}}_i^1, \ldots, \hat{\mathbf{p}}_i^S)\}_{i\in[N]}$ of the uncertain parameter and the respective probabilistic forecasts of each expert. Typically, all probability vectors are conditioned on some contextual information, with each expert potentially using a different set of features to model uncertainty; this dependency is suppressed here to simplify the notation.

2.1.2. Probabilistic forecast evaluation

Forecast quality. The merit of each expert is typically assessed based on the quality of their forecasts, through so-called proper scoring rules. A scoring rule is an extended real-valued function whose inputs are a probability distribution and an uncertainty realization. A (negatively oriented) scoring rule is proper when its expected value is minimized by the true uncertainty distribution and strictly proper when this minimizer is unique (Gneiting & Raftery, 2007). Proper scoring rules incentivize the forecaster to provide truthful predictions about the probability distribution.

Gneiting et al. (2007) posit that probabilistic forecasting aims to maximize the sharpness of the predictive distributions subject to calibration. Calibration refers to the statistical consistency between the induced probabilistic forecasts and the realizations of uncertainty, while sharpness refers to the concentration of the predictive distribution. Both calibration and sharpness can be assessed jointly through the CRPS, the state-of-the-art evaluation metric in probabilistic forecasting. Any probability vector, e.g., $\mathbf{q} \in \Sigma_K$, induces a estimation for the binary event $\{Y \leq u\}$ via its associated cumulative distribution function (CDF)

$$F(u) = \int_{-\infty}^{u} \mathbf{q}(y)dy = \sum_{y^k \leq u} q_k \tag{3}$$

and an associated quantile estimation via its associated inverse CDF $F^{-1}(\tau)$ at the level $\tau \in [0, 1]$. The CRPS evaluates the predictive CDF of $\mathbf{q}$ with a realization y of a random variable Y and is given by

$$\text{CRPS}(\mathbf{q}, y) = \int_{-\infty}^{+\infty} \big(F(u) - \mathbb{I}(y \leq u)\big)^2 du \tag{4}$$

$$= 2\int_0^1 \big(\alpha - \mathbb{I}\{y \leq F^{-1}(\alpha)\}\big)\big(y - F^{-1}(\alpha)\big)d\alpha. \tag{5}$$

The so-called quantile decomposition of CRPS (5) (Gneiting & Ranjan, 2013), shows that the CRPS has the same unit as the observation y. Even though closed-form solutions to (4) and (5) may not be available, the estimation of a discrete approximation is always feasible.

Decision quality. The decision-maker is primarily interested in the quality of decisions induced by a probabilistic forecast, as measured by the expected value of (2). A salient notion is the so-called absolute decision *regret* (Mandi et al., 2024), i.e., the excess cost incurred compared to the perfect-foresight decision. Given any $\mathbf{q} \in \Sigma_K$, let $\mathbf{z}(\mathbf{q}) = \text{argmin}_{\mathbf{z}\in\mathcal{Z}} \sum_{k\in[K]} q_k c(\mathbf{z}; \xi_k)$ be the corresponding induced decision. For a realization y_0 of uncertainty, the absolute regret, w.r.t. the cost function c and the feasible set $\mathcal{Z}$, is estimated by

$$\text{Regret}(\mathbf{q}, y_0) = c(\mathbf{z}(\mathbf{q}); y_0) - c(\mathbf{z}^*(y_0); y_0), \tag{6}$$

where $\mathbf{z}^*(y_0) = \text{argmin}_{\mathbf{z}\in\mathcal{Z}} c(\mathbf{z}; y_0)$ is the decision obtained under perfect foresight.

The regret (6) also defines a proper scoring rule relative to the distribution of Y. To see this, first note that the cost function $c(\mathbf{z}; Y)$ defines a proper scoring rule (Dawid, 2007). Let

$$\mathbf{z}(\mathbf{p}) = \operatorname*{argmin}_{\mathbf{z}\in\mathcal{Z}} \mathbb{E}_{Y\sim\mathbf{p}}[c(\mathbf{z}; Y)] = \operatorname*{argmin}_{z\in\mathcal{Z}} \sum_{k\in[K]} p_k c(\mathbf{z}, \xi_k),$$

be the optimal decision under the true distribution $\mathbf{p}$. Indeed, for any probability vector $\mathbf{q}$, it follows that

$$\mathbb{E}_{Y\sim\mathbf{p}}[c(\mathbf{z}(\mathbf{q}); Y)] \geq \mathbb{E}_{Y\sim\mathbf{p}}[c(\mathbf{z}(\mathbf{p}), Y)].$$

Further, minimizing the expected absolute regret given by

$$\min_{\mathbf{z}\in\mathcal{Z}} \sum_{k\in[K]} p_k\big(c(\mathbf{z},\xi_k) - c(\mathbf{z}^*(\xi_k),\xi_k)\big),$$

yields the same optimal solution as minimizing the expected cost, as the terms $c(\mathbf{z}^*(\xi_k),\xi_k)$, for $k \in [K]$, are constant and hence do not affect the solution. From equivalence, (6) also defines a proper scoring rule (Dawid, 2007).

Following this definition, regret is always non-negative, and an oracle with perfect foresight will incur zero regret. When $c(\mathbf{z}^*(y_0); y_0) > 0$, it is also useful to define the *relative regret*, given by $\frac{\text{Regret}(\mathbf{q},y_0)}{c(\mathbf{z}^*(y_0);y_0)}$.

2.1.3. Forecast combination

The decision-maker wants to utilize the S available forecasts to minimize their downstream costs, defined by (2) or, equivalently, the regret (6). Formally, this translates into learning a function $h : \Sigma_K \times \cdots \times \Sigma_K \to \Sigma_K$ that takes as input S probability vectors $\{\hat{\mathbf{p}}^s\}_{s=1}^S$ and combines them in a single probability vector. The decision-maker may also have access to additional contextual information, such as updated weather conditions, which can be used to improve the forecast combination. This gives rise to an augmented data set $\mathcal{D}' = \{(y_i, \hat{\mathbf{p}}_i^1, \ldots, \hat{\mathbf{p}}_i^S, \mathbf{x}_i)\}_{i\in[N]}$, where $\mathbf{x}_i$ are realizations of any additional contextual information. The goal here would be to learn a forecast combination model $h : \mathcal{X} \times \Sigma_K \times \cdots \times \Sigma_K \to \Sigma_K$ that also accounts for these additional features. We refer to this case as a *conditional* combination since it allows adapting to new contextual information and the case without additional feature data as a *static* combination.

2.2. Decision-focused linear pooling

In this section, we develop the proposed decision-focused forecast combination approach. We focus on linear pools of probabilistic forecasts; that is, we restrict the combination model h to be a linear function with non-negative coefficients that sum to one.[1] Further, we allow the linear pool weights to be a function of additional contextual information, such as current weather conditions.

The proposed decision-focused forecast combination is given by

$$\min_{f\in\mathcal{F}} \quad \sum_{i\in[N]} \text{Regret}(\hat{\mathbf{p}}_i^{\text{comb}}, y_i) + \gamma \cdot \text{CRPS}(\hat{\mathbf{p}}_i^{\text{comb}}, y_i), \tag{7a}$$

$$\text{s.t.} \quad \mathbf{z}_i^{\text{comb}} = \operatorname*{argmin}_{\mathbf{z}\in\mathcal{Z}} \sum_{k\in[K]} \hat{p}_{i,k}^{\text{comb}} c(\mathbf{z};\xi_k), \quad i \in [N], \tag{7b}$$

$$\hat{\mathbf{p}}_i^{\text{comb}} = \sum_{s\in[S]} \lambda_{i,s}\hat{\mathbf{p}}_i^s, \quad i \in [N], \tag{7c}$$

$$\boldsymbol{\lambda}_i \in \Sigma_S, \quad i \in [N], \tag{7d}$$

$$\boldsymbol{\lambda}_i = f(\mathbf{x}_i), \quad i \in [N]. \tag{7e}$$

Constraint (7b) finds the optimal decision given the combined forecast estimated from the linear pool (7c) for the ith observation. Constraint (7d) ensures a convex combination of probabilistic forecasts, and constraint (7e) is the mapping from contextual information to combination weights. The objective function (7a) minimizes a combination of the absolute decision regret, i.e., incurred decision costs against the perfect-foresight solution, and the CRPS of the induced forecast, controlled by the regularization hyperparameter γ. This hyperparameter reflects preferences concerning the potential trade-off between decision and forecast quality and can either be set by an expert or tuned via, e.g., a cross-validation strategy. As we chose to multiply the CRPS with γ, we refer to (7a) as CRPS-regularized regret. For $\gamma = 0$, we minimize the absolute decision regret only, while as $\gamma \to \infty$, we learn combination weights that minimize a discrete approximation of the CRPS for the induced forecast[2] – see, e.g., van der Meer et al. (2024). When optimizing solely for the CRPS, (7b) becomes redundant and is removed. For scaling purposes, the relative regret can be used in place of absolute regret in the objective function (7a) for strictly positive cost functions.

The function f belongs in model class $\mathcal{F} : \mathcal{X} \to \Sigma_S$. The function classes considered in this work are linear models and neural networks. Note that the output of f needs to satisfy a set of constraints, i.e., combination weights are non-negative and sum to one; a straightforward way to ensure this is by applying the softmax operator soft: $\mathbb{R}^S \to \mathbb{R}^S$, defined by $\text{soft}_S(v) = \frac{\exp(v_s)}{\sum_{j\in[S]}\exp(v_j)}$, to the output of f. For instance, a feed-forward neural network that maps features $\mathbf{x}$ to combination weights $\boldsymbol{\lambda}$ with m hidden layers and a rectified linear unit (ReLU) activation function, is given by

$$\begin{aligned}
&\mathbf{g}_0 = \mathbf{W}_0\mathbf{x} + \mathbf{b}_0,\\
&\mathbf{g}_j = \max(\mathbf{W}_j\mathbf{g}_{j-1} + \mathbf{b}_j, \mathbf{0}), && 1 \le j < m,\\
&\boldsymbol{\lambda} = \text{soft}_S(\mathbf{W}_m\mathbf{g}_{m-1} + \mathbf{b}_m),
\end{aligned}$$

where $\mathbf{W}_j$, $\mathbf{b}_j$ are the weights and bias of the jth layer and an additional softmax activation is added to the final layer to ensure constraint satisfaction. Similarly, a linear model is a special case of a neural network without hidden layers, leading to $f(\mathbf{x}) = \text{soft}_S(\mathbf{W}\mathbf{x} + \mathbf{b})$.

If the decision-maker does not have access to additional features (e.g., $\mathbf{x} = \mathbf{1}$), we consider static combination weights estimated from $f(\mathbf{x}) = \text{soft}_S(\mathbf{b})$, hence $\boldsymbol{\lambda}_i$ remains constant for all i in $[N]$. For static combinations, problem (7) (for $\gamma = 0$), provided an algorithm to solve it to optimality, finds the combination weights that lead to the lowest in-sample regret amongst any linear combination of the available component forecasts. Nonetheless, (7) is a challenging non-convex, nested optimization problem. In Section 2.3, we propose two solution methods that retain performance guarantees compared to component forecasts.

[1] While we focus exclusively on linear pooling, our approach can be adapted to more sophisticated pooling methods such as non-linear (van der Meer et al., 2024) or pointwise pooling (Berrisch & Ziel, 2023) straightforwardly.

[2] To see the equivalency, note that the objective function is currently in the form of $f(x) + \gamma g(x)$, $\gamma \ge 0$, and that scaling by a factor $\gamma + 1$ will not affect the solution. Alternatively, we can write the objective function as $\beta f(x) + (1-\beta)g(x)$, $\beta \in [0,1]$. It follows that $\gamma = \frac{1-\beta}{\beta}$ and that when $\beta \to 0$, then $\gamma \to \infty$.

Deterministic forecast combination. Further, consider the case when the downstream optimization problem (2) is modeled as an expected-value problem. Then, the decision-focused combination problem (7) simplifies considerably to learning a decision-focused linear pool of point forecasts. For LP problems with unknown cost coefficients, the regret minimization term recovers the smart "predict, then optimize" loss proposed by Elmachtoub and Grigas (2022) and the CRPS term simplifies to the mean absolute error (Gneiting & Raftery, 2007). Alternatively, the mean-squared error can be used as a regularizer; in that case, as $\gamma \to \infty$, we converge to the method of Granger and Ramanathan (1984).

2.3. Solution methodology

This section develops two solution methods for problem (7) based on differential optimization layers and in-sample performance weighting.

2.3.1. Differential optimization

We leverage the fact that the solution map of convex optimization problems is differentiable and propose a first-order, gradient-based method to approximate (7), akin to the method proposed by Agrawal et al. (2021).

Let $\boldsymbol{\theta}$ be a vector that concatenates the model parameters of f. We use stochastic gradient descent to learn $\boldsymbol{\theta}$, as detailed in Algorithm 1. In the forward pass, we sample a batch of data, linearly combine the expert forecasts, solve the respective optimization problems, and estimate the CRPS-regularized regret (7a). In each backward pass, we update model parameters using the gradient of the CRPS-regularized regret (7a) w.r.t. $\boldsymbol{\theta}$. This involves estimating (i) the gradient of cost w.r.t. the decisions $\mathbf{z}$ induced by the forecast combination, (ii) the gradient of decisions $\mathbf{z}$ w.r.t. the combination weights $\boldsymbol{\lambda}$, and (iii) the gradient of $\boldsymbol{\lambda}$ w.r.t. the model parameters $\boldsymbol{\theta}$.

We now discuss some important points regarding the gradient-based algorithm. Firstly, evaluating the gradient of the decisions $\mathbf{z}$ w.r.t. combination weights $\boldsymbol{\lambda}$, which requires differentiating through the argmin operator of (7b). This is implemented through implicit differentiation of the Karush Kuhn Tucker (K.K.T.) equations (Amos & Kolter, 2017), which comes at a low computational cost. If the cost function $c(\mathbf{z}; Y)$ is linear, it may be difficult to obtain useful gradients. To alleviate this problem, we introduce a regularization term $\alpha \cdot \|\mathbf{z}\|_2^2$ in (7b), where α is typically small enough (e.g., 0.001) so that it does not affect the solution much. Secondly, each forward pass requires solving N optimization problems. While the algorithm is meant to run offline, we note the potentially high computational costs. An alternative approach that reduces this computational cost is developed in Section 2.3.2. Note that the computational cost of the forward pass is not affected by the number of experts S; hence, we do not expect a significant cost increase when S is high. Thirdly, we note that the composition of the softmax operator with f in the forward pass ensures that $\boldsymbol{\lambda} \in \Sigma_S$; hence, we do not need to implement a projection step in the backward pass.

Algorithm 1 Gradient-based learning

Input: Training data set $\mathcal{D} = \{(y_i, \hat{\mathbf{p}}_i^1, \dots, \hat{\mathbf{p}}_i^S)\}_{i\in[N]}$, set of hyperparameters (learning rate η, batch size β, total number of epochs)
Output: Model parameters $\boldsymbol{\theta}$.

1: Initialize model parameters $\boldsymbol{\theta}$.
2: Sample a batch of data $\mathcal{B} = \{(y_i, \hat{\mathbf{p}}_i^1, \dots, \hat{\mathbf{p}}_i^S)\}_{i\in[\beta]}$.
3: Linearly pool component forecasts, solve the inner maximization problems (7b) and estimate the CRPS-regularized regret.
4: Update model parameters

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \frac{\eta}{|\mathcal{B}|} \nabla_{\boldsymbol{\theta}} \sum_{i\in[b]} \text{Regret}(\hat{\mathbf{p}}_i^{\text{comb}}, y_i) + \gamma \cdot \text{CRPS}(\hat{\mathbf{p}}_i^{\text{comb}}, y_i).$$

5: Repeat Steps 2-4 till a stopping criterion is met.

The iterative process described in Algorithm 1 is repeated until a stopping criterion is met. In our implementation, we set an upper limit of iterations over the training data set (epochs) and run the algorithm till the solution stops improving, which signifies convergence to a (local) optimum. When learning static combination weights (i.e., f is constant), we initialize the gradient descent algorithm multiple times and select the best-performing ones. By initializing $\boldsymbol{\lambda} = (0, 0, \dots, 1, 0, \dots, 0)$ where 1 occupies the position corresponding to the expert with the lowest in-sample regret, we guarantee that in-sample CRPS-regularized regret is upper-bounded by the best-performing expert. Similarly, we can initialize $\boldsymbol{\lambda}$ from other linear combinations, e.g., an ordinary linear pool, and strictly improve upon them in terms of regret. We use the standard learning approach of assessing performance on a hold-out validation set for conditional combinations (i.e., f is a linear or nonlinear model). The details of selecting the respective hyparameters are discussed in Section 3.1.2.

2.3.2. Performance-based inverse weighting

The main bottleneck of the gradient-based approach is solving multiple optimization problems at each forward pass, and computational costs can become significant when training for multiple epochs. To this end, for the case of static combinations, we propose weighing individual forecasts based on historical performance, adapting the well-known weighting scheme of Bates and Granger (1969) in a decision-focused framework.

The process is described as follows. First, for each expert s and each observation i, we solve the contextual stochastic optimization problem (2) ($S \times N$ problems in total). Next, for each expert s, we estimate the average in-sample CRPS-regularized regret, denoted by v_s. Finally, the static combination weights for the sth expert are given by

$$\lambda_s = \frac{\frac{1}{v_s}}{\sum_{s\in[S]} \frac{1}{v_s}}, \tag{8}$$

i.e., by inversely weighing the in-sample performance of each expert. Similar to the gradient-based approach,

for $\gamma \to \infty$, the experts are weighted based on their in-sample CRPS.

3. Numerical experiments

This section presents the numerical experiments. Section 3.1 describes the experimental setup. Section 3.2 provides an illustrative example with synthetic data. Section 3.3 presents the results of the renewable trading case study. Section 3.4 presents the results of the grid scheduling case study. The code to recreate the experiments is available on GitHub.[3]

3.1. Experimental setup

3.1.1. Component forecasts

For individual component forecasts, we utilize non-parametric machine learning models, such as tree-based ensembles, which achieve state-of-the-art performance in contextual stochastic optimization problems (Bertsimas & Kallus, 2020) with minimal tuning effort. Specifically, each expert has access to a training data set $\{y_i, \mathbf{x}_i\}_{i\in[N_{\text{train}}]}$ of joint observations of uncertainty and contextual information. Note that each expert may have access to a different feature vector $\mathbf{x}_i$; we detail how $\mathbf{x}_i$ is constructed separately for each case study. The proposed models then learn a function that assigns weights $\boldsymbol{\omega}(\mathbf{x}) \in \Sigma_N$ to training observations y_i based on a new realization of contextual information $\mathbf{x}_0$. These weighted historical observations comprise the probabilistic forecast. The original contextual stochastic optimization problem (2) can then be approximated by solving a weighted sample average approximation problem, given by

$$\min_{\mathbf{z}\in\mathcal{Z}} \sum_{i\in[N]} \omega_i(\mathbf{x}_0) c(\mathbf{z}; y_i). \tag{9}$$

As Y has finite support, we can count the number of times ξ_k appears in $\mathcal{D}$ and aggregate the respective weights $\omega_i(\mathbf{x}_0)$ to equivalently write (9) with a probability vector that weighs each support location. That is, the estimated probability of $Y = \xi_k$ conditioned on $\mathbf{x}_0$ is given by $\sum_{i\in[N]} \mathbb{I}(y_i = \xi_k)\, \omega_i(\mathbf{x}_0)$. The following models are used as experts:

1. **Nearest Neighbors**: We consider weights learned with a k Nearest Neighbors (kNN) model (Hastie, Tibshirani, & Friedman, 2001, Chap. 13), given by
$$\omega_i(\mathbf{x}_0) = \frac{1}{k}\mathbb{I}(\mathbf{x}_i \text{ is a neighbor of } \mathbf{x}_0),$$
where hyperparameter k indicates the number of data points closest to $\mathbf{x}_0$.
2. **Decision Trees**: We consider weights learned with decision tree models using the CART algorithm (Breiman, Friedman, Stone, & Olshen, 1984). Let $\tau : \mathcal{X} \to \{1, \dots, L\}$ be a map that corresponds to a disjoint partition of $\mathcal{X}$ into L tree leaves and $\tau(\mathbf{x}_0)$ is the identity of the leaf that $\mathbf{x}_0$ falls into. The respective weights are given by
$$\omega_i(\mathbf{x}_0) = \frac{\mathbb{I}(\tau(\mathbf{x}_i) = \tau(\mathbf{x}_0))}{\sum_{i=1}^{N} \mathbb{I}(\tau(\mathbf{x}_i) = \tau(\mathbf{x}_0))}.$$
3. **Tree-based Ensembles**: We consider weights learned with an ensemble of decision trees, grown with the Random Forest (RF) algorithm (Breiman, 2001). Consider an ensemble of T decision trees $\{\tau_1, \dots, \tau_T\}$, where $\tau_j : \mathcal{X} \to \{1, \dots, L_j\}$ is a map that corresponds to a disjoint partition of $\mathcal{X}$ into L_j tree leaves and $\tau_j(\mathbf{x}_0)$ is the leaf identity. The respective weights are given by
$$\omega_i(\mathbf{x}_0) = \frac{1}{T}\sum_{j=1}^{T} \frac{\mathbb{I}\left(\tau_j(\mathbf{x}_i) = \tau_j(\mathbf{x}_0)\right)}{\sum_{i'=1}^{N} \mathbb{I}\left(\tau_j(\mathbf{x}_{i'}) = \tau_j(\mathbf{x}_0)\right)}.$$

3.1.2. Combination methods

The following methods of estimating the weights of a linear pool are compared:

1. **Ordinary Linear Pool** (`OLP`): We assign uniform weights to all the experts (Stone, 1961), a robust approach that performs well in practice.
2. **CRPS learning** (`CRPSL`): We learn combination weights by minimizing the CRPS of the combined forecast, similar to the objective considered by Berrisch and Ziel (2023), Thorey et al. (2017), van der Meer et al. (2024).
3. **Decision-focused learning** (`DFL-`γ): We learn combination weights by solving the decision-focused combination problem (7) using the gradient-based approach outlined in Section 2.3.1 for different values of hyperparameter γ.
4. **Performance-based inverse weighting** (`invW-`γ): Each expert is weighted inversely to their historical performance in terms of CRPS-regularized regret, as proposed in (8), for different hyperparameter values γ.

The numerical experiments examine the potential trade-off between decision and forecast quality and how it evolves as a function of γ. To this end, we consider a grid of values $\gamma = \{0, 0.1, 1, \infty\}$ and train multiple variations of `DFL-`γ, `invW-`γ. Note that `DFL-`∞ is equivalent to `CRPSL` and hence omitted.

In the case of the conditional forecast combination using additional features, we consider two classes of models $\mathcal{F}$ for the function f that adapts the combination weights, namely linear regression (`LR`) and neural network-based (`NN`) models. Conditional forecast combination is only tested for `CRPSL` and `DFL-`γ. For clarity, `DFL-LR-`γ is learned by solving the decision-focused combination problem (7) where f is a linear model, and so forth.

All variants of `DFL-`γ and `CRPSL` are optimized with a stochastic gradient-based approach,[4] using the Adam algorithm (Kingma & Ba, 2014) with a batch size of 500, a learning rate of 0.01. Modeling is implemented in `PyTorch` (Paszke et al., 2019) and the differentiable optimization layers are implemented with the `cvxpylayers` (Agrawal et al., 2019) package. For the static

[3] https://github.com/akylasstrat/df-forecast-comb.

[4] As Y is discrete, `CRPSL` can be cast as a convex optimization problem and solved to optimality. Nonetheless, the gradient-based approach offers computational benefits when the number of training observations is large, which is why we apply it in our numerical experiments.

combination methods (CRPSL, DFL-γ), we iterate till convergence. For the conditional combination methods (CRPSL-LR, CRPSL-NN, DFL-LR-γ, DFL-NN-γ), we evaluate performance using a subset of the training set as a validation set. We set the total number of epochs high and train using early stopping, i.e., we stop training if the validation performance fails to improve for 25 epochs. For the NN-based conditional combinations, we consider a feedforward network with 3 hidden layers and 20 nodes per layer.

3.1.3. Performance evaluation

We evaluate the efficacy of the different combination methods w.r.t. decision and forecast quality, using a test set of N^{test} observations. Decision quality is assessed by estimating the average out-of-sample regret

$$\frac{1}{N^{\text{test}}} \sum_{i \in [N^{\text{test}}]} c(\mathbf{z}_i^{\text{comb}}; y_i) - c(\mathbf{z}_i^*(y_i); y_i),$$

where $\mathbf{z}_i^{\text{comb}}$ is the decision obtained by each forecast combination method and $\mathbf{z}_i^*(y_i)$ is the perfect-foresight decision. Perfect foresight implies knowledge of the realization y_i and, by design, leads to zero regret. Forecast quality is measured by approximating the average CRPS over the test set using its quantile decomposition (5) for a grid of $\{0.01, 0.02, \ldots, 0.99\}$ of quantiles.

3.2. Synthetic data example

Problem description. We first showcase the efficacy of the proposed combination methods in an illustrative example with synthetic data. We consider an uncertain variable Y associated with features X_0, X_1, X_2, X_3, which all follow a standard normal distribution, through

$$Y = X_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 \cdot \mathbb{I}(X_3 \leq \beta) + \epsilon, \tag{10}$$

where ϵ denotes an error term and $\alpha_1 = \alpha_2 = 1.2$, $\alpha_3 = 4.5$, and $\beta = -1.3$ are known parameters. Two experts use the available features to generate probabilistic forecasts. Features X_1 and X_2 denote information measured only by expert 1, X_3 denotes information measured only by expert 2, while X_0 denotes publicly available information. The two experts issue the following conditional normal distributions:

$$p_1 = \mathcal{N}(X_0 + \alpha_1 X_1 + \alpha_2 X_2, 1),$$
$$p_2 = \mathcal{N}(X_0 + \alpha_3 X_3 \cdot \mathbb{I}(X_3 \leq \beta), 1).$$

The decision-maker uses these two component forecasts to minimize the newsvendor loss (Ban & Rudin, 2019)

$$c(z; Y) = \max\Big(\frac{\tau}{1-\tau}(Y - z), (z - Y)\Big),$$

where τ is the optimal quantile.

Assume that the optimal quantile is set at $\tau = 0.20$, which targets the left tail of the distribution. By design, expert 1 does a better job, on average, modeling the conditional distribution of Y. In comparison, expert 2 does a relatively better job modeling the distribution's left tail (i.e., lower quantiles). We expect a higher weight to expert 2 in the linear pool to lead to lower decision costs. We validate this assumption by sampling 10 000 observations with 50/50 training/test split and evaluating the different combination methods.

Table 1
Results for synthetic data example. Bold font indicates the best-performing combination method.

	λ_1	λ_2	Regret	CRPS
Expert 1	1	0	68.437	8.995
Expert 2	0	1	51.565	10.110
OLP	0.500	0.500	26.842	6.539
invW-0	0.419	0.581	26.282	6.709
invW-0.1	0.419	0.581	26.282	6.708
invW-1	0.423	0.577	26.482	6.697
invW-∞	0.521	0.479	26.817	6.521
CRPSL	0.532	0.468	27.003	6.516
DFL-0	0.409	0.591	**26.146**	6.741
DFL-0.1	0.500	0.500	26.842	6.539
DFL-1	0.536	0.464	27.003	**6.514**

Results. Table 1 presents average out-of-sample regret and CRPS (scaled values) alongside the static combination weights $\boldsymbol{\lambda}$ learned by each method. Concerning the component forecasts, expert 1 leads to higher forecast quality (lower CRPS), and expert 2 leads to higher decision quality (lower regret), as expected. All combination methods significantly improve upon the component forecasts in both evaluation metrics. DFL-0 ranks first in terms of decision quality, followed by invW-0, invW-0.1, while DFL-1 and CRPSL are almost equivalent and the best in terms of forecast quality.

Interestingly, CRPSL is the worst-performing combination method in terms of decision quality, while DFL-0 is the worst-performing one in terms of forecast quality, which is attributed to the learned combination weights. As shown from the quantile decomposition in (5), CRPS can be approximated by the quantile score, hence, CRPSL weighs experts' performance equally across the whole distribution. As expert 1 does a good job modeling a larger part of the distribution, CRPSL assigns her a larger weight, with $\lambda_1 = 0.532$. Conversely, DFL-0 only considers the quantile loss at the 20-th quantile, i.e., the newsvendor loss for $\tau = 0.20$; as expert 2 does a better job modeling the left tail of the uncertainty distribution, DFL-0 assigns her higher weight, with $\lambda_2 = 0.591$. As expected, as γ increases, the weights learned by DFL-γ converge to those of CRSPL, indicating the trade-off between forecast and decision quality. Similar behavior is observed for invW-γ, although the learned combination weights remain closer to the average value of 0.5. Namely, invW-0 assigns a weight of $\lambda_2 = 0.581$ to expert 2, which is slightly smaller than the respective weight learned by DFL-0; similarly, invW-∞ are closer to the ones learned by CRPSL.

3.3. Solar production forecasting and trading in electricity markets

Problem description. We consider a renewable producer participating in a day-ahead electricity market as a price-taker. This problem is pivotal for integrating renewable energy sources in modern power systems and electricity markets (Morales et al., 2013). Before market closure, typically around 12:00 of the day $d-1$, the producer submits an energy offer in the day-ahead market for each hourly clearing period of day d. During real-time operation, the system operator activates balancing reserves to maintain

the demand-supply equilibrium and stabilize the system frequency. In the case of real-time production shortage (demand exceeds supply), the operator activates upward reserves, while downward reserves are activated in the case of a production surplus (supply exceeds demand). Market participants who deviated from their day-ahead contracts recover the cost of balancing reserve activation. This means that, based on real-time production, a renewable producer has to buy back (sell) the amount of energy shortage (surplus) to balance their position.

We consider a dual-price balancing mechanism, which means that production deviations that help stabilize system frequency are not penalized, e.g., overproducing when the system faces a shortage. Conversely, producers whose deviations aggravate the system imbalance settle their positions at the marginal cost of reserves, incurring a financial loss. The problem of deriving an optimal trading offer can be formulated as a newsvendor problem (Morales et al., 2013) [Ch. 7]. Here, we consider a hybrid trading strategy proposed by Stratigakos et al. (2022) where the producer balances trading costs and point forecast accuracy, hedging potential trading risks. The producer's cost function is given by

$$c(z;Y) = (1-\rho)\max\Big(\frac{\tau}{1-\tau}(Y-z),(z-Y)\Big) + \rho(Y-z)^2, \tag{11}$$

i.e., a combination of the newsvendor and mean squared error loss, where Y is the uncertain production for a clearing period, z is the energy offer, ρ is a design parameter that controls the degree of risk aversion, and τ is the optimal quantile, which is estimated by the expected difference between day-ahead prices and marginal regulation costs. Since prices are unknown, τ is estimated based on historical observations and typically takes values smaller than 0.50, as upward reserves tend to be costlier. Further, Y is normalized by the nominal plant capacity, taking values in the discrete set $\{0, 0.01, \ldots, 0.99, 1\}$, and the feasible decision set is $\mathcal{Z} = \{0 \leq z \leq 1\}$.

Data and component forecasts. We use approximately 2.5 years of hourly production data for three solar plants located in Australia, provided by the Global Energy Forecasting Competition 2014 (GEFCom2014) (Hong et al., 2016). The contextual information comprises weather forecasts from a Numerical Weather Prediction (NWP) model obtained in a day-ahead horizon. The NWP model outputs 12 weather variables, including precipitation, solar radiation, temperature, relative humidity, cloud cover, and wind components — see Hong et al. (2016) for details. We use one year of data to train individual forecast experts, one year of data to estimate the combination weights and seven months as a test set. The NWP forecasts are issued at 6:00 of the day $d-1$ and cover every hour of the day d, i.e., a forecast horizon of 18-42 h ahead, which mirrors the real-world participation of renewable producers in day-ahead electricity markets. We generate component forecasts using the *k*NN, CART, and RF models described in Section 3.1.1. We perform a grid search with 5-fold cross-validation for hyperparameter tuning and re-train using the whole training data set once the hyperparameters are selected. We further consider an additional set of 4 features that model diurnal patterns, using Fourier terms that model daily and hourly seasonality patterns. We assume these are available only to the decision-maker and use them to train the conditional forecast combination methods.

Results. Fig. 1 shows the probabilistic forecast performance of each expert. Although RF is the best-performing component forecast overall, as confirmed by its CRPS (normalized by nominal plant capacity) in Table 2, Fig. 1 shows that performance varies across the distribution regions. Namely, CART performs better for the middle part of the distribution, while *k*NN performs well for its lower tail and considerably worse for its right tail.

We evaluate decision quality for $\rho = 0.2$, $\tau = \{0.1, 0.2, \ldots, 0.9\}$, and each solar plant in the data set. Table 2 shows the average out-of-sample regret and CRPS (both scaled) for all combination methods for a single solar farm and $\rho = 0.2, \tau = 0.2$, alongside learned combination weights $\boldsymbol{\lambda}$ (for static combinations). Results for the different values of τ and the rest of the solar farms are similar and thus omitted.

We first examine static forecast combinations, indicated with a white background color in Table 2. Overall, all combination methods significantly improve upon component forecasts both in terms of decision and forecast quality. RF is the best-performing component forecast in both metrics. DFL−0.1 leads to the highest forecast quality (lowest CRPS), which is approximately 10% better than RF, closely followed by DFL−1 and CRPSL. Conversely, DFL−0 leads to the highest decision quality, with an average regret of approximately 12.6% lower than RF. Indeed, the results of Table 2 highlight that higher forecast quality does not guarantee better decisions. Concerning invW−γ, all variations outperform OLP but are worse than both DFL−γ and CRPSL in both metrics, while OLP is the worst-performing combination method. We next examine the variability of learned combination weights $\boldsymbol{\lambda}$ across the static combination methods. Namely, CRPSL assigns a very small weight to *k*NN ($\lambda_1 = 0.039$), while the respective weight for CART is much larger ($\lambda_2 = 0.535$); interestingly, although RF is the best-performing component forecast in terms of CRPS, it does not receive the highest weight in the linear pool. Conversely, the weights learned by DFL−0 are more evenly distributed, with the respective weight of *k*NN being 0.188 and CART being 0.387, while RF receives the highest weight of 0.425. As γ increases, we observe that the respective weights of the DFL−γ method move closer to the weights learned by CRPSL, as expected. Conversely, the weights learned by invW−γ are very close to the ones of OLP, which potentially explains why it fails to outperform either of CRPSL and DFL−γ.

Next, we examine performance for conditional forecast combinations, highlighted with gray in Table 2. Overall, conditional forecast combinations improve upon static combinations in terms of decision and forecast quality, which is expected as the former utilizes additional feature data. The average improvement, however, is less than 1% in most cases, primarily attributed to the experiment design and the nature of the feature data. Indeed, the

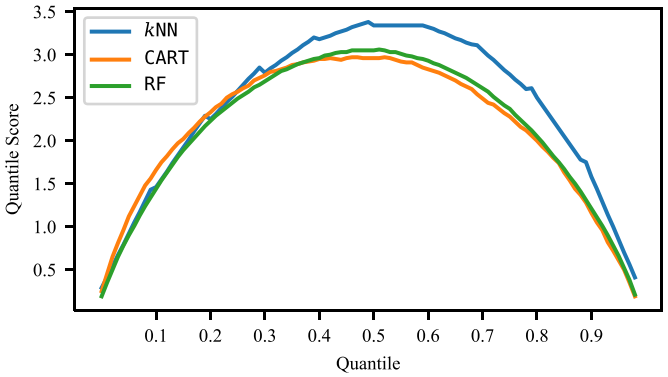


Fig. 1. Solar production forecasting: Quantile score (scaled) evaluated at an evenly spaced grid of quantiles $\{0.01, 0.02, \ldots, 0.99\}$.

Table 2
Learned combination weights $\boldsymbol{\lambda}$, average hourly decision regret, and CRPS for a single solar plant ($\tau = 0.2$, $\rho = 0.2$). Bold font indicates the best-performing method for static and conditional combinations, respectively. Underlined bold font indicates the best-performing method overall.

		λ_1	λ_2	λ_3	Regret	CRPS
Static combinations	kNN	1	0	0	2.663	5.412
	CART	0	1	0	2.653	5.075
	RF	0	0	1	2.578	5.049
	OLP	0.333	0.333	0.333	2.282	4.596
	invW-0	0.322	0.338	0.341	2.279	4.590
	invW-0.1	0.320	0.340	0.341	2.278	4.589
	invW-1	0.314	0.346	0.340	2.275	4.583
	invW-∞	0.310	0.351	0.339	2.274	4.580
	CRPSL	0.039	0.535	0.426	2.263	4.523
	DFL-0	0.188	0.387	0.425	**2.251**	4.544
	DFL-0.1	0.083	0.529	0.388	2.258	**4.521**
	DFL-1	0.078	0.535	0.387	2.260	4.522
Conditional combinations	CRPSL-LR	–	–	–	2.259	4.427
	CRPSL-NN	–	–	–	2.251	**4.425**
	DFL-LR-0	–	–	–	2.236	4.527
	DFL-NN-0	–	–	–	**2.229**	4.485
	DFL-LR-0.1	–	–	–	2.259	4.426
	DFL-NN-0.1	–	–	–	2.246	4.439
	DFL-LR-1	–	–	–	2.257	4.426
	DFL-NN-1	–	–	–	2.249	4.440

NWP variables already capture the diurnal effect through solar radiation forecasts; hence, the additional features leveraged by the decision-maker provide small benefits. The differences across learning models are also small, with NN leading to slightly better performance in most cases. CRSPL-NN ranks first in terms of forecast quality (lowest CRPS), while DFL-NN-0 ranks first in terms of decision quality (lowest regret).

Finally, Fig. 2 presents the average percentage improvement of CRPL, DFL$-\gamma$, and their conditional variations, over OLP across all the experiments, namely the three solar plants and the different values of optimal quantile τ. A key takeaway is that optimizing the weights of the linear pool improves over OLP. Notably, optimizing for a combination between regret and CRPS ($\gamma > 0$) in the decision-focused combination (7) consistently leads to great performance in both metrics. For the static combination case, DFL-1 outperforms CRSPL and DFL-0 in both metrics. For the conditional combination case, DFL-0.1 and DFL-1 lead to a lower CRPS compared CRPSL, while also improving in terms of regret. DFL-1 leads to the lowest CRPS overall across all experiments. DFL-0 leads to consistent regret improvements, which become more pronounced when considering conditional combinations. These results corroborate other works that posit that optimizing decision cost and forecast accuracy leads to the best trade-off.

3.4. Wind production forecasting and grid scheduling

Problem description. We further consider a realistic grid scheduling problem under net demand uncertainty. A grid operator schedules the production level of a set of G generators to meet an uncertain future net demand Y, defined as base electricity demand minus renewable production. We assume the operator schedules production in a day-ahead horizon (18-42 h ahead), consistent with the market-clearing and scheduling processes currently implemented in Europe and the USA. The operator utilizes probabilistic forecasts to anticipate potential regulation costs due to a real-time mismatch between demand and supply, which is critical for future low-carbon power

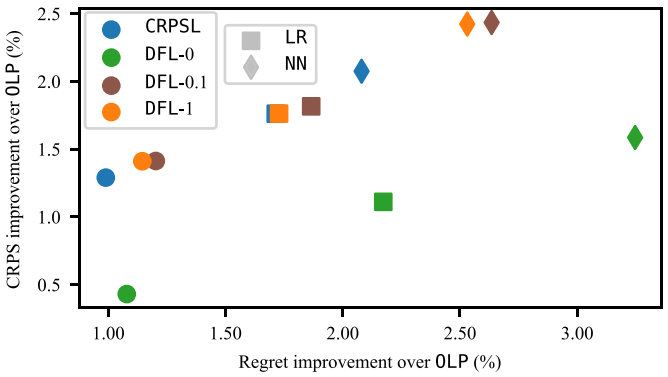


Fig. 2. Regret improvement versus CRPS improvement against OLP, average value over the 3 solar plants and $\tau = \{0.1, 0.2, \ldots, 0.9\}$. The circle marker indicates static combinations. The square and diamond markers indicate conditional combinations using linear and neural network models, respectively. Values toward up and right are better.

systems (Hobbs et al., 2022). The grid scheduling problem is formulated as a two-stage stochastic optimization problem with relatively complete recourse given by

$$\min_{\mathbf{z},\mathbf{z}_k^u,\mathbf{z}_k^d} \quad \mathbf{c}^\top \mathbf{z} + \sum_{k\in[K]} p_k \left(\mathbf{c}^{u\top}\mathbf{z}_k^u - \mathbf{c}^{d\top}\mathbf{z}_k^d\right), \tag{12a}$$

$$\text{s.t.} \quad \mathbf{1}^\top(\mathbf{z} + \mathbf{z}_k^u - \mathbf{z}_k^d) = \xi_k, \qquad k \in [K], \tag{12b}$$

$$\mathbf{0} \le \mathbf{z} \le \overline{\mathbf{z}}, \tag{12c}$$

$$\mathbf{0} \le \mathbf{z}_k^u \le \min(\overline{\mathbf{z}}^u, \overline{\mathbf{z}} - \mathbf{z}), \qquad k \in [K], \tag{12d}$$

$$\mathbf{0} \le \mathbf{z}_k^d \le \min(\overline{\mathbf{z}}^d, \mathbf{z}), \qquad k \in [K]. \tag{12e}$$

where $\mathbf{z}$ is the vector of look-ahead (first-stage) generator dispatch decisions, $\{\mathbf{z}_k^u, \mathbf{z}_k^d\}$ is the set of recourse (second-stage) actions, namely upward and downward regulation, defined per each scenario k, $\overline{(\cdot)}$ indicates upper limits on decision variables, and $\mathbf{c}^d \le \mathbf{c} \le \mathbf{c}^u$ (inequality is applied elementwise) are non-negative cost vectors. Namely, $\mathbf{c}$ denotes linear generation costs, $\mathbf{c}^d$ denotes downward regulation costs, and $\mathbf{c}^u$ denotes upward regulation costs. The objective function (12a) minimizes the expected dispatch cost under net demand uncertainty considering real-time recourse actions, (12b) ensures the demand-supply balance under all possible realizations of uncertainty modeled as discrete scenarios, and constraints (12c)–(12e) are the technical constraints of the generators, namely, upper and lower production limits (12c) and ramping constraints for upward (12d) and downward regulation (12e), defined per each scenario. Here, the min operator is applied elementwise. For scaling purposes, we use the relative regret when optimizing the linear pooling weights by dividing (12a) with the cost of the perfect-foresight solution, which does not incur any regulation costs. The scaled uncertainty takes values in the grid $\{0, 0.01, \ldots, 0.99, 1\}$. Finally, we included a small regularization term in all DFL-γ variants to improve the convergence of the gradient-based algorithm.

Data and component forecasts. We consider a scheduling problem where a constant base demand is coupled with a wind farm, and two thermal generators are used to meet the resultant stochastic net demand. Thermal generator data are obtained from the IEEE 14-bus system (Babaeinejadsarookolaee et al., 2019), with upward regulation costs $\mathbf{c}^u$ sampled from a range $[1.1 \cdot \mathbf{c}, 6 \cdot \mathbf{c}]$ and downward regulation costs $\mathbf{c}^d$ sampled from a range $[0.05 \cdot \mathbf{c}, 0.2 \cdot \mathbf{c}]$. We assume a constant demand of 200 MW, a wind farm of 100 MW nominal capacity, and appropriately scaled generator capacity data to the same aggregate capacity value. We use two years of hourly wind production data provided by GEFCom2014 (Hong et al., 2016) and use one year of data to train expert models, six months to estimate combination weights and six months for out-of-sample testing. As in the previous case study, the contextual information comprises weather forecasts from an NWP model obtained in a day-ahead horizon: zonal and meridional wind components at 10 m and 100 m hub height. We further process these to generate wind speed and wind direction forecasts at the respective hub heights (4 features in total), as is standard in wind production forecasting. The NWP forecasts are issued at 6:00 of the day $d-1$ and are used for scheduling the production of the day d, with a horizon of 18-42 h ahead, as in real-world market-clearing and scheduling processes. We generate component forecasts using the kNN, CART, and RF models as experts— see Section 3.1.1, and the wind production data from zone 1 as the target. All experts utilize the same weather variables as input features. To create variability in the component forecasts, each expert utilizes NWP features from different locations of the GEFCom2014 data set. As before, we perform a grid search with 5-fold cross-validation for hyperparameter tuning and re-train using the whole training data set once the hyperparameters are selected.

Results. Fig. 3 plots the forecast quality of component forecasts. CART consistently outperforms kNN, RF across the distribution, which is attributed to the sampling of NWP features, as the ones utilized by CART are closer to the target wind farm location. This differs from the previous case study in Section 3.3, where experts' performance varied across the distribution.

Table 3 presents the average out-of-sample regret (in 10^3 \$ per hour) and CRPS (normalized by nominal plant capacity) over the test set, alongside the learned combination weights $\boldsymbol{\lambda}$ for static combination methods. To estimate the regret for the ith test observation, which marks

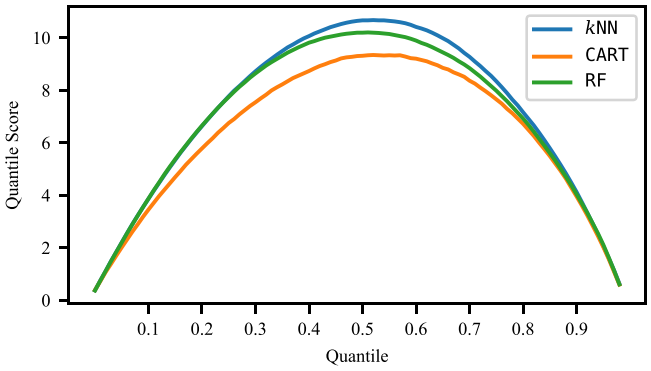


Fig. 3. Wind production forecasting: Quantile score (scaled) evaluated at an evenly spaced grid of quantiles $\{0.01, 0.02, \ldots, 0.99\}$.

Table 3
Learned combination weights $\boldsymbol{\lambda}$, average hourly regret (10^3\$/h), and CRPS for a grid scheduling problem with 100 MW wind plant and 2 thermal generators. Bold font indicates the best-performing combination method.

	λ_1	λ_2	λ_3	Regret ($\frac{10^3\$}{h}$)	CRPS
kNN	1	0	0	570.995	7.821
CART	0	1	0	427.126	6.328
RF	0	0	1	575.322	7.646
OLP	0.333	0.333	0.333	438.320	6.448
invW-0	0.328	0.346	0.327	436.631	6.422
invW-0.1	0.328	0.346	0.327	436.631	6.422
invW-1	0.328	0.346	0.327	436.631	6.422
invW-∞	0.323	0.358	0.319	435.013	6.399
CRPSL	0.324	0.626	0.050	409.747	6.089
DFL-0	0.131	0.765	0.104	**385.628**	**6.078**
DFL-0.1	0.277	0.613	0.110	407.505	6.088
DFL-1	0.300	0.618	0.081	408.506	6.089

an hour of operation, we first solve the stochastic scheduling problem (12) to find the look-ahead dispatch actions. Once uncertainty y_i is realized, we fix the look-ahead decision variables and solve the scheduling problem (12) again to find the least-cost recourse actions that maintain the demand-supply balance during real-time operation. The final cost of the ith observation is then estimated as the sum of the look-ahead and regulation costs minus the perfect foresight cost (clearly, perfect foresight requires no real-time regulation).

From Table 3, we observe that CART outperforms all the component forecasts in both metrics. Interestingly, CART also outperforms OLP and all invW-γ variations, which have almost evenly distributed weights. Conversely, both CRPSL and DFL-γ variants improve upon both CART, with DFL-0 ranking first in both metrics. Concerning forecast quality, CRPSL and DFL-γ perform similarly, while more notable differences are observed for regret. Namely, DFL-0 leads to 9.7% lower regret and 3.9% lower CRPS compared to CART; in terms of regret, DFL-0 is 5.8% better than CRSPL. Regarding the weight distribution, both DFL-γ and CRPSL recognize CART as the most important expert and assign it the largest weight. Compared to CRPSL, DFL-0 assigns a higher weight to both CART and RF, while also assigning significantly smaller weight to kNN.

4. Conclusions

This work proposed a novel approach for combining probabilistic forecasts that account for downstream decision costs. We developed decision-focused linear pooling, where combination weights are optimized to minimize an expected cost function of a stochastic optimization problem. We extended the proposed method to the case where the combination weights adapt to the realization of additional features available to the decision-maker. A comprehensive evaluation was conducted considering synthetic and real-world test cases, examining integral problems associated with renewable energy integration in low-carbon power and energy systems. The results highlighted the proposed decision-focused combination approach's efficacy, which improved ordinary linear pooling with uniform weights in decision and forecast quality. In particular, the average reduction in decision regret compared to ordinary linear pooling was approximately 2% in a trading problem with stochastic solar production and 12% in a grid scheduling problem with stochastic wind production. Notably, optimizing for CRPS-regularized regret consistently led to a better trade-off between decision and forecast quality. Overall, this work highlighted the benefits of embedding the downstream objective when combining forecasts, which allowed us to maintain a simple setting of linear pooling while significantly improving decision quality.

There are several interesting avenues to explore in future work. First, this work can be extended for the case of multivariate uncertainty, e.g., by using the Energy Score (Gneiting & Raftery, 2007), a proper scoring rule for multivariate probabilistic forecasts, as a regularizer instead of the CRPS. Second, speeding up the learning process will be useful to enable large-scale applications; one interesting idea is to utilize machine learning-based optimization proxies to reduce the computational burden. Finally, exploring decision-focused approaches for other combination methods, such as non-linear pooling and quantile averaging, is interesting.

CRediT authorship contribution statement

Akylas Stratigakos: Writing – review & editing, Writing – original draft, Visualization, Validation, Software,

Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Salvador Pineda:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Juan Miguel Morales:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The code and data necessary to reproduce the results of IJF-D-24-00254R2 are available at the following Github repository: https://github.com/akylasstrat/df-forecast-comb.

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